

Recursive Bayesian Estimation for Cross-Sectional Momentum States

Umang Duttatreya

How can we transform static, backward-looking cross-sectional momentum (Information Discreteness) into a forward-looking Bayesian state filter to accurately isolate alpha across shifting market regimes?

Inspiration: The FIP Momentum

The Frog-in-the-Pan (FIP) Hypothesis

The Behavioral Premise:

- *Investors possess limited attention capacity.*
- *Markets adequately price large, sudden information shocks (Discrete Signals).*
- *Markets systematically underreact to small, continuous information arrivals (Continuous Signals).*

Much like the old fable of a frog that does not notice water slowly heating up until it is too late, investors fail to pay close attention to continuous, incremental information flows.

The Quantitative Proxy: Information Discreteness (ID)

Information Discreteness (ID): Measures how evenly or discretely information flows. Continuous information (steady trickle) causes a delayed but strong, persistent price continuation (momentum) that doesn't quickly reverse. Dramatic, one-off news (discrete jumps) attracts immediate attention and gets priced in instantly.

Our benchmark ID proxy is determined by the sign of daily returns and ignores their magnitude by equally weighting each observed return. The percentage of days during the formation period with positive and negative returns are denoted $\%pos$ and $\%neg$, respectively.⁶ ID is defined as

$$ID = \text{sgn}(\text{PRET}) \times [\%neg - \%pos], \quad (1)$$

where the cumulative return during the formation period is denoted PRET. Specifically, PRET is defined as a firm's cumulative return over the past twelve months after skipping the most recent month. The sign of PRET is denoted $\text{sgn}(\text{PRET})$ and equals: +1 when $\text{PRET} > 0$ and -1 when $\text{PRET} < 0$.

Gaps in Traditional FIP Momentum vs. Proposed Stochastic Framework

Traditional FIP Limitations	Proposed Bayesian-State Upgrades
Static Calendar Lookback: Relies on rigid 252-day windows; treats day 1 and day 252 with equal weight, ignoring recent structural fatigue.	Recursive Stochastic Filter: Continuously updates state parameters (α , β) with exponential decay (λ) to weight recent information flow dynamically.
Cold-Start Bias & Arbitrary Priors: Assumes uninformative or uniform starting distributions, causing early-window instability.	Empirical Bayes Priors (MLE): Calibrates initial parameters ($\alpha_0 = 128.5$, $\beta_0 = 117.7$) from cross-sectional market data to provide statistical shrinkage and stability.
Static Return Magnitudes: Assumes historical return sizes remain fixed over time, ignoring shifting volatility regimes.	Dynamic Likelihood Updating: Recursively updates conditional expectations (μ_A, μ_B) using incoming daily sample likelihoods.

The Idea

Observation: Traditional FIP momentum relies entirely on tracking binary daily outcomes (Up vs. Down days) to measure trend continuity.

Hypothesis: Since FIP is fundamentally just counting discrete binary events, it maps perfectly to a probability distribution. Instead of a static accounting ratio, could we model historical trend continuity as a Prior Distribution, updating it dynamically as each new trading day closes?

$$\text{Posterior} \propto \text{Likelihood} \times \text{Prior}$$

The Trading Application: By shifting from a static rank to a continuous probability state, we can calculate a forward-looking **Expected Return ($E[R_{t+1}]$)** for every stock in the cross-section to directly drive our daily systematic trading decisions.

Methodology

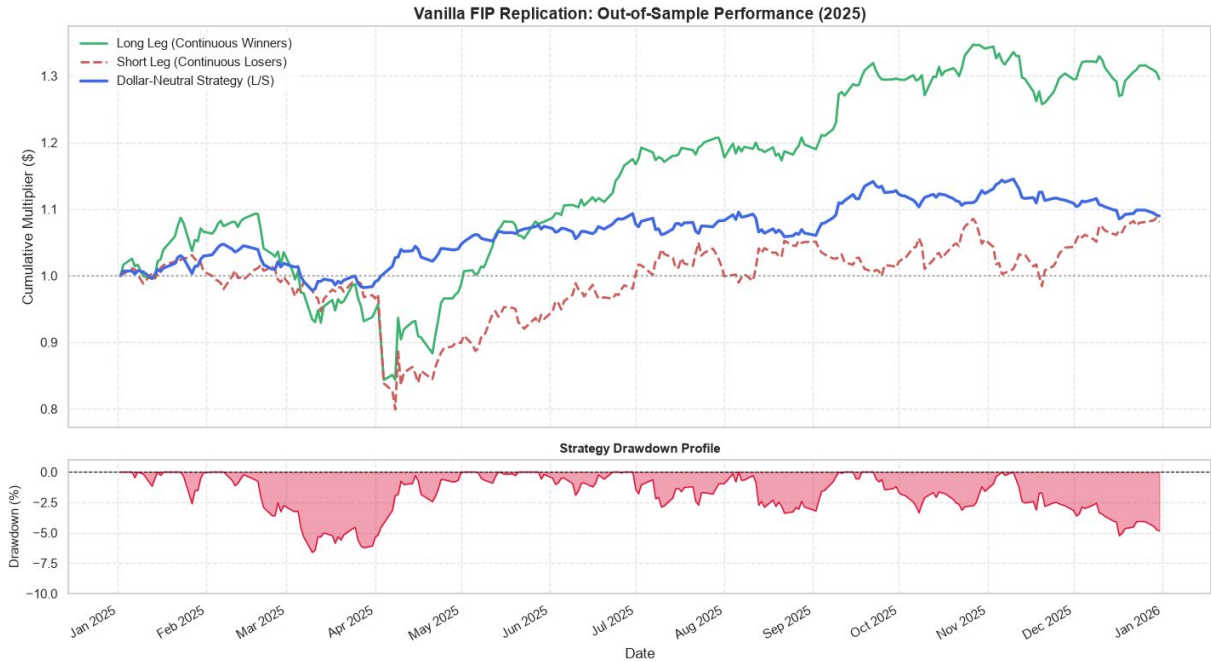
—

Data Overview & Experimental Setup

1. **Dataset & Universe:** Daily OHLC pricing data sourced from Kaggle, covering a broad cross-section of distinct stocks. Adjusted Close prices utilized to ensure total return accuracy (incorporating stock splits and dividends).
2. **In-Sample / Training Set (2024):** Used exclusively for Empirical Bayes prior calibration (estimating α_0, β_0 via MLE) and fitting conditional return expectation distributions.
3. **Out-of-Sample / Test Set (2025):** Used solely for sequential, step-by-step backtesting of the dynamic Bayesian filter.
4. **Data Hygiene & Preprocessing:** Filtered out missing records and zero-volume/illiquid trading days to prevent pricing anomalies. Calculated simple daily returns: $R_t = P_t / P_{t-1} - 1$. **Strict temporal separation enforced to guarantee zero lookahead bias or data leakage.**

Baseline Benchmark: Static Vanilla FIP (2025 Out-of-Sample)

Metric	Static 1 Year Baseline
Annualized Return	8.97%
Annualized Volatility	9.49%
Sharpe Ratio	0.94
Max Drawdown	-6.61%



Takeaways

1. **Proof of Concept:** A standalone Sharpe of 0.94 on a frictionless factor indicates that Information Discreteness (ID) isolates a true market underreaction signal.
2. **Drawdown Containment:** The sub-7% maximum drawdown demonstrates that the dependent double-sort successfully neutralizes broader market beta.
3. **Structural Limitation:** A static 1-year holding period causes capital stagnation, holding decaying trends too long without capturing real-time trend acceleration or breakdown.

Next, we construct backtests for our hypothesis.

Empirical Prior Calibration (2024 In-Sample)

To initialize the Bayesian filter without arbitrary assumptions, we extract two distinct sets of empirical priors from the 2024 cross-section:

1. The Directional Prior (Beta-Bernoulli Conjugate)

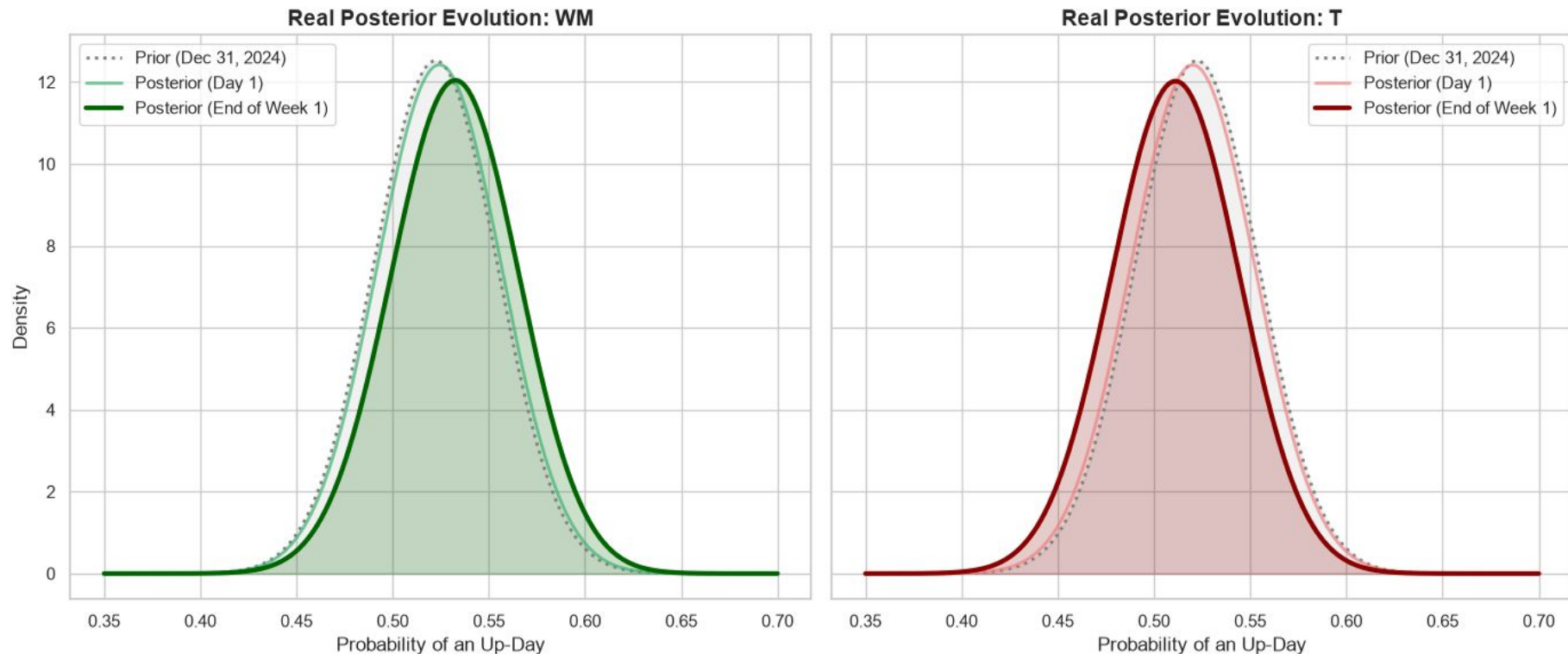
- Models the probability of an **up-day vs. a down-day**.
- Calibrated via **Maximum Likelihood Estimation (MLE)** across the 2024 universe to establish the global starting states ($\alpha_0 = 128.5, \beta_0 = 117.7$).

2. The Magnitude Prior (Conditional Expectations)

- Models the payoff size given a specific directional outcome.
- Calculated as the empirical mean of positive returns (R_A) and negative returns (R_B) for each individual ticker.

$$E[R_{t+1}] = P(\text{Up})_t \cdot R_{A,t} + P(\text{Down})_t \cdot R_{B,t}$$

Inside the Black Box: Dynamic Posterior Evolution



The exponential decay factor ($\lambda = 0.98$) ensures the model rapidly discounts stale 2024 information, allowing 2025 micro-regimes to pull the probability mass toward the new reality within just 5 trading days.

The Decay Factor (λ)

1. The Role of Memory (λ):

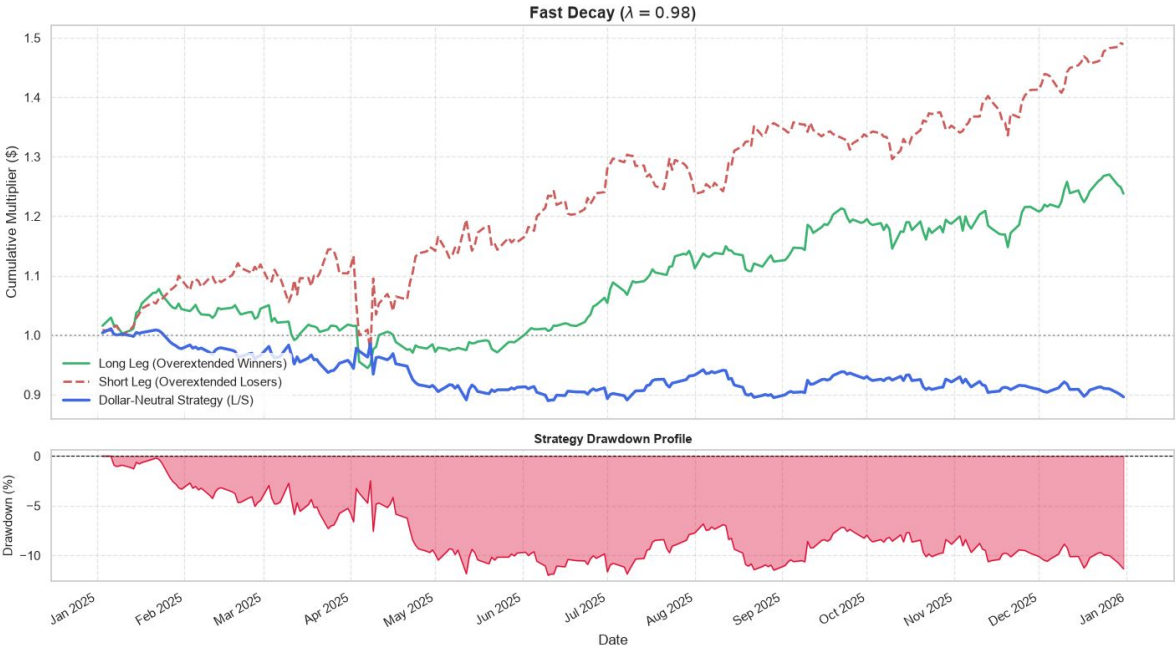
λ dictates the filter's half-life: how quickly historical priors are overwritten by new information.

$$\text{Half-Life} \approx \ln(0.5)/\ln(\lambda)$$

2. **The Discovery (The Failed Backtest):** Setting $\lambda = 0.98$ (roughly a 34-day half-life) resulted in a **Sharpe Ratio of -0.80** and a downward equity curve.
3. **The Diagnosis:** The model did not fail; it successfully identified a different market anomaly. By speeding up the memory, the filter crossed the boundary from medium-term momentum into short-term mean reversion.

Backtest 1: Momentum Strategy in a Mean Reversion Regime

Metric	Value
Annualized Return	-10.40%
Annualized Volatility	13.02%
Sharpe Ratio	-0.80
Max Drawdown	-11.99%



Cumulative out-of-sample performance using a fast decay factor ($\lambda = 0.98$). The consistent downward drift in the dollar-neutral spread highlights the structural boundary where medium-term FIP momentum decays into short-term mean reversion.

Tuning λ for Dual Alpha Generation

1. Signal 1:

a. Capturing Momentum Calibration:

- i. Slow $\lambda = 0.998$ (roughly a 346-day half-life).
- ii. **Logic:** The filter ignores 2-week noise, smoothly tracking the 12-month underreaction premium.
- iii. **Execution:** Buy Continuous Winners, Short Continuous Losers.

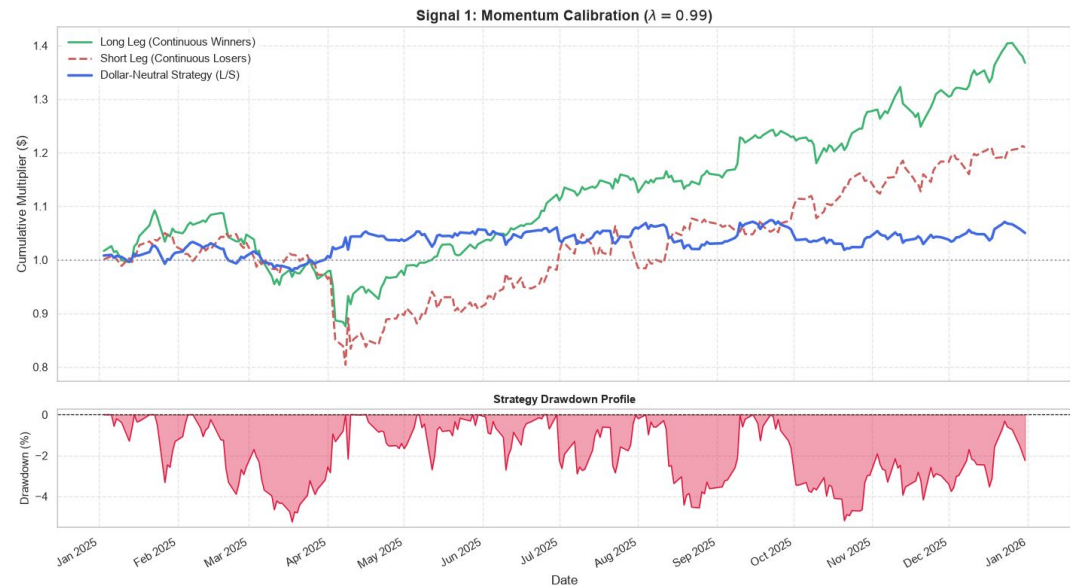
2. Signal 2:

a. Harvesting Short-Term Reversal Calibration:

- i. Fast $\lambda = 0.98$ (roughly a 34-day half-life).
- ii. **Logic:** The filter aggressively tracks immediate, short-term trend exhaustion.
- iii. **Execution:** Invert the FIP targets. Buy Continuous Losers, Short Continuous Winners.

Backtest 2: Momentum Calibration

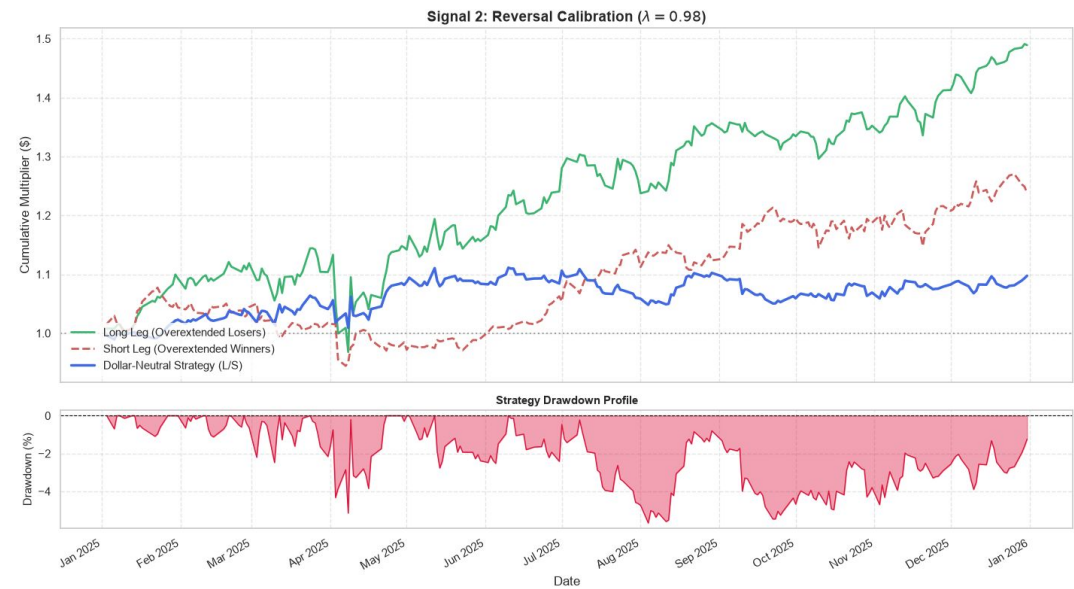
Metric	Value
Annualized Return	5.01%
Annualized Volatility	11.36%
Sharpe Ratio	0.44
Max Drawdown	-5.25%



Out-of-sample performance for the corrected medium-term momentum calibration ($\lambda = 0.99$). The Bayesian filter successfully captures the structural Information Discreteness premium, generating a stable, uncorrelated return stream with strictly bounded drawdowns.

Backtest 3: Reversal Calibration

Metric	Value
Annualized Return	7.53%
Annualized Volatility	12.76%
Sharpe Ratio	0.59
Max Drawdown	-7.96%



Cumulative out-of-sample performance of the inverted strategy execution. Maintaining a fast memory ($\lambda = 0.98$) while flipping the dependent sort targets allows the Bayesian framework to successfully exploit short-term price snaps and mean reversion.

Conclusions

1. **Dynamic State Translation:** The Bayesian framework shows a promising transformation from the static, backward-looking Information Discreteness (ID) accounting ratio into a dynamic, forward-looking probability state.
2. **Dual Alpha Generation:** The exponential decay factor (λ) acts as a regime-switching dial; tuning this single memory parameter allows the exact same mathematical framework to extract two orthogonal alpha streams.
3. **Risk Containment:** Both the momentum and reversal calibrations demonstrated highly defensive properties out-of-sample, successfully neutralizing broader market beta and capping maximum drawdowns at -5.25% and -7.96%, respectively.

Limitations

1. **Limited Backtest Horizon:** The current out-of-sample validation is restricted to a single calendar year (2025). This limited window does not expose the filter to a complete macroeconomic cycle or systemic black-swan events.
2. **Frictionless Assumptions:** The metrics for the fast-decay reversal calibration ($\lambda = 0.98$) do not explicitly account for elevated transaction costs and slippage inherent in high-turnover execution.
3. **Magnitude Shock Sensitivity:** Relying on historical mean distributions for return magnitudes (R_A, R_B) may underestimate the impact of sudden, fat-tailed discrete earnings gaps.

Further Research & Algorithmic Evolution

1. **Rigorous Parameter Calibration:** Upgrading the manual decay selection by deploying Monte Carlo simulations and empirical cross-validation to algorithmically calibrate the optimal λ half-life across different volatility regimes.
2. **Individual Priors:** Transitioning from a uniform global prior distribution ($\alpha_0 = 128.5, \beta_0 = 117.7$) to individual, stock-specific priors to ensure the baseline probabilities accurately reflect unique asset-level historical trends.
3. **High-Frequency Intraday Translation:** Translating the Beta-Bernoulli conjugate prior framework from daily OHLC data to intraday tick data to model low-latency order flow continuity.



Thank You